

0 math voodoo

$$\ln x \leq x - 1.$$
$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots, \quad x \in \mathbb{R}.$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad |x| < 1.$$

Markov's inequality: For nonnegative X ,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0.$$

Chebyshev's inequality:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}, \quad a > 0.$$

Jensen's inequality: For convex f ,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

1 Measures

1.1 Information

To quantify the **amount of information** in an event occurring with probability p , we find a function ψ such that:

- $\psi(p)$ is **non-negative**.
- $\psi(1) = 0$. Definite events carry no information.
- $\psi(p)$ is **continuous**, and decreasing in p . Unlikely events carry more information.
- $\psi(p_1 p_2) = \psi(p_1) + \psi(p_2)$. Independent events carry additive information.

The only function that satisfies these axioms is, for some base b :

$$\psi(p) = \log_b \frac{1}{p} \quad p \in [0, 1].$$



We consider $b = 2$ to measure information in **bits**. Any other base is equivalent up to a constant factor.

1.2 Shannon Entropy

Consider a discrete random variable X with PMF $p(x)$. To quantify the **entropy** of (the distribution of) X such that:

- $H(\mathbf{p})$ is **continuous** as a function of $\mathbf{p}$.
- $H(\mathbf{p})$ is **maximised for the uniform distribution**, and is increasing in the number of outcomes.
- Considering successive decisions, the following holds:

$$H(\mathbf{p}) = H(p_1 + p_2, \dots, p_N) + (p_1 + p_2) H_2\left(\frac{p_1}{p_1 + p_2}, \frac{p_2}{p_1 + p_2}\right)$$

The only function that satisfies these axioms is the **Shannon entropy**, up to a constant factor:

$$H(P_X) = \mathbb{E}_{X \sim P_X} \left[\log_2 \frac{1}{P_X(X)} \right].$$

Entropy is a measure of **uncertainty/information** in X .

Binary entropy: In particular, for a Bernoulli distribution with probability p , the entropy is given by the **binary entropy function**:

$$H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p).$$

Uniform distribution: For a uniform distribution over N outcomes, the entropy is given by:

$$H(\mathbf{p}) = \log_2 N.$$

Entropy generalises to **joint** and **conditional distributions**:

$$H(X, Y) = \mathbb{E}_{(X, Y) \sim P_{X, Y}} \left[\log_2 \frac{1}{P_{X, Y}(X, Y)} \right],$$

$$H(X|Y) = \mathbb{E}_{(X, Y) \sim P_{X, Y}} \left[\log_2 \frac{1}{P_{X|Y}(X|Y)} \right].$$

$H(X|Y)$ can be interpreted as the **uncertainty in X after observing Y** .

Properties of Shannon entropy

- $H(X) \geq 0$. Entropy is **non-negative**, and is zero iff X is deterministic.
- $H(X) \leq \log_2 |\mathcal{X}|$, with equality iff X is uniform.
- Chain rule**:

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$$

- $H(X|Y) \leq H(X)$, with equality iff X, Y are independent. **NOTE**: $H(X|Y = y)$ could exceed $H(X)$.
- $H(X_1, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$, with equality iff $X_1, \dots, X_n$ are independent.
- Data processing inequality**: For any deterministic function f , $H(X) \geq H(f(X))$.
- Information-preserving transformations**: For any bijection f , or if Y depends on X only through $f(X)$, $H(X) = H(f(X))$.

- Concavity**: $H(\lambda P + (1-\lambda)Q) \geq \lambda H(P) + (1-\lambda)H(Q)$

1.3 KL Divergence

To quantify the **difference between two distributions P and Q** , we use the **KL Divergence**:

$$D_{\text{KL}}(P||Q) = \mathbb{E}_{X \sim P} \left[\log_2 \frac{P(X)}{Q(X)} \right].$$

- $D(P||Q)$ is **non-negative**, and is zero iff $P = Q$.
- Chain Rule**: Let $C := D(P_{Y|X} || Q_{Y|X} | P_X)$. $D(P_{XY} || Q_{XY}) = D(P_X || Q_X) + C$, $C = \sum_x P_X(x) D(P_{Y|X}(\cdot|x) || Q_{Y|X}(\cdot|x))$
- Data processing inequality**: For any function f , $D(P_X || Q_X) \geq D(P_{f(X)} || Q_{f(X)})$.

1.4 Mutual Information

To quantify the **"information gain"** of observing another variable, we use the **mutual information**:

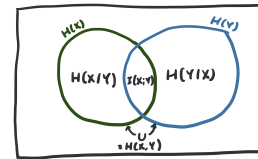
$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$
$$I(X; Y|Z) = H(X|Z) - H(X|Y, Z) = H(Y|Z) - H(Y|X, Z)$$

and **joint distributions**:

$$I(X; Y, Z) = H(X) - H(X|Y, Z).$$

Alternative forms:

$$I(X; Y) = D(P_{XY} || P_X P_Y)$$
$$= \mathbb{E} \left[\log_2 \frac{P_{XY}(X, Y)}{P_X(X) P_Y(Y)} \right]$$
$$= \mathbb{E} \left[\log_2 \frac{P_{Y|X}(Y|X)}{P_Y(Y)} \right]$$



Properties of mutual information

- Non-negativity**: $I(X; Y) \geq 0$, with equality iff X and Y are independent.
- $I(X; Y) \leq \min \{H(X), H(Y)\}$ are **upper bounds** on information gain.
- $I(X; Y) = H(X)$ iff X is a **deterministic** function of Y .
- Symmetry**: $I(X; Y) = I(Y; X)$.
- Chain rule**:

$$I(X_1, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_1, \dots, X_{i-1})$$

- Data processing inequality**: If $X \rightarrow Y \rightarrow Z$ forms a Markov chain, then

$$I(X; Z) \leq I(X; Y).$$

- Partial sub-additivity**: If $Y_1, \dots, Y_n$ are conditionally independent given $X_1, \dots, X_n$, and Y_i depends on $X_1, \dots, X_n$ only through X_i , then

$$I(X_1, \dots, X_n; Y_1, \dots, Y_n) \leq \sum_{i=1}^n I(X_i; Y_i).$$

- Convexity**: $I(X; Y)$ is convex in $P_{Y|X}$ for fixed P_X , and concave in P_X for fixed $P_{Y|X}$.

2 Symbol-wise Source Coding

Setup

We denote the set of all symbols (the **alphabet**) by $\mathcal{X}$. Consider a discrete (memoryless) source X with PMF P_X over $\mathcal{X}$.

Lowercase English text: $\mathcal{X} = \{a, \dots, z\}$, with $P_X(e)$ being the highest.

We encode each string $x \in \mathcal{X}^*$ into a decodeable binary string $C(x)$, of length $\ell(x)$. The goal is to minimize the **expected codeword length** of the code C :

$$L(C) = \mathbb{E}_{X \sim P_X} [\ell(X)] = \sum_{x \in \mathcal{X}} P_X(x) \cdot \ell(x),$$

2.1 Symbol-Wise Coding

Symbol-wise coding encodes **each symbol** $x \in \mathcal{X}$ into a codeword $C(x)$, and **encodes a string** $x_1 \dots x_n$ by concatenating the codewords $C(x_1) \dots C(x_n)$.

Decodeability Conditions

Non-singular codes: A code is non-singular if all codewords are distinct.

$$\forall x, x' \in \mathcal{X}, x \neq x' \implies C(x) \neq C(x').$$

Uniquely decodable codes: A code is uniquely decodable if any concatenation of codewords can be uniquely parsed into codewords.

That is, for distinct $x_1 \dots x_n, x'_1 \dots x'_m \in \mathcal{X}^*$,

$$C(x_1) \dots C(x_n) \neq C(x'_1) \dots C(x'_m).$$

Uniquely decodeability is **non-trivial to verify**.

Prefix-free codes: A code is prefix-free if no codeword is a prefix of another codeword.

C is prefix-free $\implies C$ is uniquely decodable.

Additionally, prefix-free codes can be decoded **instantaneously**, without needing to wait for the entire string to be received.

Kraft's inequality

Kraft's inequality states that for any prefix-free code C ,

$$\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1.$$

Any prefix-free code can be **represented as a binary tree**, where each codeword corresponds to a leaf. The depth of the leaf is the codeword length. It follows, by random traversal of the tree, that the **probability of reaching a leaf** is $2^{-\ell(x)}$.

Conversely, for any set of codeword lengths $\{\ell(x)\}_{x \in \mathcal{X}}$ satisfying Kraft's inequality, there exists a prefix-free code with these codeword lengths.

Entropy lower bound

Entropy provides a fundamental **limit on the expected codeword length** of any uniquely decodable code:

$$L(C) \geq H(X),$$

with equality iff $P_X(x) = 2^{-\ell(x)}$ for all $x \in \mathcal{X}$.

Using the entropy lower bound, **Shannon-Fano coding** is a simple method to construct a **prefix-free code for a given distribution P_X** .

Ordering the symbols in decreasing probability, define:

$$C(x) = \text{binary expansion of } \sum_{x' < x} P_X(x'),$$

truncated to $\ell(x) = \lceil -\log_2 P_X(x) \rceil$ bits.

A Shannon-Fano coding satisfies the **expected codeword length** of $H(X) \leq L(C_{SF}) \leq H(X) + 1$.

Mismatched coding: If the true distribution is P_X but **lengths are chosen according to Q_X** , the Shannon-Fano code has expected codeword length

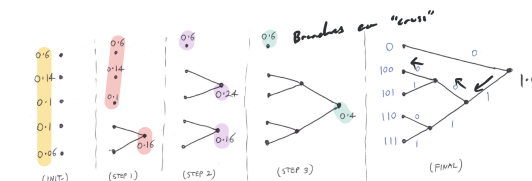
$$L(C_Q) = H(P_X) + D(P_X || Q_X).$$

2.2 Huffman Coding

Huffman coding constructs an **optimal prefix-free code** for a given distribution P_X . The **algorithm** is as follows:

- Start with a forest of $|\mathcal{X}|$ trees, each consisting of a single node corresponding to a symbol $x \in \mathcal{X}$, with weight $P_X(x)$.
- While there are more than one tree in the forest:
 - Select the two roots with the smallest weights, say T_1 and T_2 .
 - Merge T_1 and T_2 into a new tree T , where T_1 and T_2 are the left and right subtrees of T , respectively. The weight of T is the sum of the weights of T_1 and T_2 .
- The remaining tree in the forest is the Huffman tree.

At each node, assign a binary digit to each branch. The code for a given symbol is the **sequence of binary digits along the path** from the root to the leaf corresponding to that symbol.



Optimality: The expected codeword length of the Huffman code is

$$L(C_H) = \min_{C \text{ prefix-free}} L(C).$$

2.3 Extensions of Symbol-wise Coding

Block coding: Instead of encoding each symbol separately, we can **encode blocks of n symbols** together. We consider instead the **alphabet $\mathcal{X}^n$** , and the distribution P_{X^n} .

This allows us to achieve expected codeword lengths closer to the entropy, as n increases. $\forall \epsilon > 0$, there exists a block code C s.t.:

$$L(C) < H(X) + \epsilon.$$

Advantages of block coding:

- Encode sources with memory, by treating blocks as symbols.
 $P(\text{apple}) > P(\text{ealpp})$ despite containing the same characters.
- Achieve expected codeword length arbitrarily close to the entropy. Even with independent symbols,

$$NH(X) \leq L(C) \leq NH(X) + 1.$$

$$H(X) \leq \text{Average length per symbol} \leq H(X) + \frac{1}{N}.$$

Disadvantages of block coding:

- Determining the product distribution P_{X^n} can be difficult, especially for sources with memory.
- Encoding and decoding complexity grows exponentially with n . Requires sorting $|\mathcal{X}|^n$ symbols by probability, and storing the codebook of size $|\mathcal{X}|^n$.